A Federated Learning Method for Real-time Emotion State Classification from Multi-modal Streaming

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ABSTRACT

Emotional and physical health are strongly connected and should be taken care of simultaneously to ensure completely healthy persons. A person's emotional health can be determined by detecting emotional states from various physiological measurements (EDA, RB, EEG, etc.). Affective Computing has become the field of interest, which uses software and hardware to detect emotional states. In the IoT era, wearable sensor-based real-time multi-modal emotion state classification has become one of the hottest topics. In such setting, a data stream is generated from wearable-sensor devices, data accessibility is restricted to those devices only and usually a high data generation rate should be processed to achieve real-time emotion state responses. Additionally, protecting the users' data privacy makes the processing of such data even more challenging. Traditional classifiers have limitations to achieve high accuracy of emotional state detection under demanding requirements of decentralized data and protecting users' privacy of sensitive information as such classifiers need to see all data. Here comes the federated learning, whose main idea is to create a global classifier without accessing the users' local data. Therefore, we have developed a federated learning framework for real-time emotion state classification using multi-modal physiological data streams from wearable sensors, called Fed-ReMECS. The main findings of our Fed-ReMECS framework are the development of an efficient and scalable real-time emotion classification system from distributed multimodal physiological data streams, where the global classifier is built without accessing (privacy protection) the users' data in an IoT environment. The experimental study is conducted using the popularly used multi-modal benchmark DEAP dataset for emotion classification. The results show the effectiveness of our developed approach in terms of accuracy, efficiency, scalability and users' data privacy protection.

1. Introduction

The link between emotion and health has long been a subject of scientific investigation in psychology, but it took on new significance with the introduction of Alexander's psychosomatic paradigm [**?**]. Emotion plays a big role in our thinking process and behavioural activities. It affects our psychological states and therefore our bodies which directly implies to our health [**?**]. Because what is psychological is ultimately biological. For example, poor emotional health can weaken the body's immune systems, making us more vulnerable to infection during emotionally difficult times. In addition to that, when we do not feel good, our emotional state is negative; we may not take care of our health (may not feel like eating, taking medicine, etc.). So, it is evident that detecting the emotional states from physiological measurements is necessary to ensure that our emotional health is well, which implies good health.

With the recent advancement in Internet of Things (IoT) technologies, we have easy access to those physiological measuring (EDA, ECG, etc.) devices that are wearable in our day-to-day lives. Furthermore, the rapid growth of mobile devices opens up new opportunities to extract emotional states more frequently by using those mobile devices for physiological measurements, allowing us to properly care for our emotional health, which translates to good health. So, to take care of our emotional health daily, there is a need for wearable-sensor-based real-time emotion classification. Because real-time emotion classification from mobile wearable-sensor devices has the following properties:

1. Decentralised data: generated data comes from mobile wearable-sensor devices of users.
2. Data have sensitive information such as location data, health information and online identifiers (IP address) etc.
3. Data have velocity because the high data generation rate and sequential arrival in time, according to data streams. The arrived data tuple needs to be processed and analysed as soon as they arrived because real-time emotion state classification.

Now, when it comes to handle decentralised data along with safeguarding the privacy and sensitive information of users, federated learning (a distributed learning approach [**?**]) is best because this allows users to collaboratively benefit from shared models, trained on their local data without transferring data to others. The federated learning approach fulfils the first two above mentioned properties. In order to fulfil the last the three properties the supervised data streaming approaches (such as Real-time Multi-modal Emotion Classification System called ReMECS [**?**]) is required. Having said that, to solve the whole problem, we have developed a federated learning method for real-time emotion state classification using multi-modal physiological data streams

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from wearable sensors, Fed-ReMECS, which fulfills the aforementioned demanding requirements.

Fed-ReMECS is made up of a central server and a number of mobile users who are all running the same ReMECS application. All users' model weights are collected by the server. It assigns average weights to each user for model updates, and no user ever uploads raw data created locally to the server. Keeping in mind the IoT part and mobile devices, Message Queuing Telemetry Transport (MQTT) protocol is used because it is bandwidth-efficient, lightweight and uses little battery power. Also, it is a *publish/subscribe* messaging protocol. Which means the publisher publishes (send) the messages to a topic (which means a context) and the subscriber subscribes (reads) from a particular topic. Our main focus here is to develop an efficient and scalable real-time emotion classification system from distributed multimodal physiological data streams, where the global classifier is built without accessing (privacy protection) the users' data in an IoT environment.

The main research contributions of this paper are as follows:

1. We design a federated learning method for real-time emotion from multi-modal data streams, named Fed-ReMECS. The Fed-ReMECS consists of a central server and multiple mobile users who all are running our previously developed ReMECS application. The whole approach is made suitable for IoT environment by using MQTT a publisher/subscriber lightweight protocol.
2. The Fed-ReMECS is evaluated using the popular benchmark multi-modal dataset DEAP for emotion classification. The evaluated result shows the effectiveness of our proposed method in terms of accuracy.

The rest of the paper is structured as follows: section 2 introduces to the brief ideas of multi-modal data stream classification, federate learning and emotion state representation. The material and methods for Fed-ReMECS is presented in section 3. In section 4, the experiment results and discussion is provided. The real-life application scenarios of our proposed approach is described in section 5. Finally the paper ends with the conclusion in section 6.

2. Preliminaries

Herewith, a brief introduction to the learning from the multi-modal data streams, federated learning and relevant emotional models are presented.

2.1. Multi-modal Data Streams Classification

The term *modality* refers to sensory modalities, which are the major means of communication and feeling in humans. The term "*multi-modal*" refers to the use of various modalities or sensor modalities to complete a task. Data are generated continually and often sent in real-time as small data tuples (Kilobytes). The multi-modal data stream now refers to unbounded data tuples from various sensor modalities that are continuous and unbounded. The model will perform badly at start because to a lack of appropriate comprehension of the data, but it will progressively learn and update itself from the data stream, finally improving its classification performance. The simplified progressive learning of the stream classifier consists of following steps [?]:

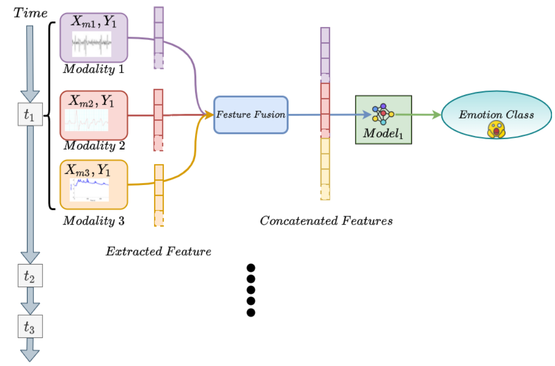

Figure 1: Emotion classification from multi-modal data stream in online mode.

• Receive the data tuple or instance (x_t) without its actual label.

• Make the class prediction $\hat{y}_t = f_t(x_t)$ for x_t, where f is the current model.

• Receive the true class label y_t for x_t immediately after the classification.

• Train the model f_t using the pair (x_t, y_t) and update the model's performance metrics using $(\hat{y}_t, y_t)$.

• Proceed to the next data tuple or instance.

Now, in multi-modal data stream classification scenario, features are extracted from the arrived data tuples from each modality and then a feature fusion approach (feature concatenation [?]) is applied and then stream classifier is used to classify emotion classes which follows the above mentioned steps for their progressive learning. Here, the time dimension (t), shown in Fig. 1, is important to note, where different data tuples arrive in a stream mode at different time (t_1, t_2, …) from different modality.

In multi-modal data streams classification scenario, the following assumptions are made:

• Data tuples are arriving sequentially one at a time.

• The model first receives the unlabeled data tuples of the stream and then *immediately* after making the prediction the *true class labels* arrives.

• Model is tested first and then trained (interleaved-test-then-train approach).

• Model can see the data tuple at most once (Multiple run through data is not possible).

• There is only a temporary storage for data stream, means no looped back over received data.

2.2. Emotion State Representation

Defining multiple emotional states in a meaningful ways is very challenging task in emotion studies and it's been a debate in psychology [?]. In literature, emotions are represented in different continuous or distinct levels [?] (e.g., valence-arousal-dominance (VAD) or pleasant-unpleasant-attention-rejection). The VAD representation is the most popularly used in emotion recognition research. The valence measure provides the measure of pleasantness ranging from negative to positive, arousal measure provides the degree of excitability ranging from high to low and dominance measure provides the degree of control that a person has over his emotion states. According to [**?**], the dominance measure is very hard to quantify and that leads to a 2D emotion representation model called Valence-Arousal (VA) model, which is also known as Russell's 2D model. Having said that, the Russells 2D model can also be mapped to categorical or discrete emotion labels such as happy, sad, tired etc. In [**?**], authors have shown the mapping between the valence-arousal 2D emotion space to categorical or discrete emotion labels. Having the valence-arousal measurements are very crucial in the mapping but the discrete or categorical emotion labels are easier to understand than the valence-arousal measures.

2.3. Federated Learning

Federated Learning (FedL [**? ?**]) is a machine learning scenario in which the aim is to train a high-quality centralized model using training data spread over a large number of clients, each having unreliable and relatively network connections. The training data is saved locally on users' mobile devices, and the devices act as nodes, doing computation on their local data to update a global model. By bringing model training to the device, this goes beyond the usage of local models that generate predictions on mobile devices. The FedL framework is not the same as traditional distributed machine learning because of the enormous number of clients, extremely imbalanced and non-i.i.d. data accessible on each client, and very weak network connections.

There are three major components in an FedL: clients, the server, and the aggregation framework to prepare the high quality global model.

Clients. FedL's clients are its shareholders. During the training session, they have access to local data. They are exclusively responsible for calculating the local model and submitting their own parameters to the server.

Server. The server is typically a strong computing node. It manages the communication between clients and itself to train the global machine learning model based on the received local model parameters.

Aggregation Framework. The aggregation framework in FedL is in charge of managing two tasks: computation and communication operations. Computation occurs on both the client and the server, while communication occurs between the participants and the management. Typically, computing is used for model training, and communication is used to exchange model parameters. Federated Averaging is a simple and extensively used framework (FedAvg) [**?**].

Table 1
Brief description of DEAP dataset

Description	Details
Total participants	32 (16 male-16 female)
Experimented content	40 music videos of length 60s each
Physiological signals available	EEG, EDA, RB
Available channels	Total of 47 channels (32 channels for EEG, 12 channels of peripheral, 3 unused)
Physiological recordings	Available in .dat file of total 32 participants
Data representation	3D matrix representation $40 \times 40 \times 8064$, which represents video/trial $\times$ channel $\times$ data
Label representation	2D matrix representation (40×4)
Available emotion labels	valence, arousal, dominance and liking

3. Materials and Methods

In this section the considered benchmark dataset for emotion classification, experimental study, experimental setup and considered performance metrics for performance evaluation is briefly discussed.

3.1. Data Set Description

The effectiveness of the proposed approach is evaluated using the most popular and widely used benchmark dataset DEAP [**?**] for emotion recognition from multi-modal physiological signals. This dataset provide a diverse range of subjects (human) on which the experiment is conducted and huge variety of different physiological signal measurements. The brief description of DEAP dataset is as follows.

DEAP dataset[1]*:* DEAP [**?**] (Database for Emotion Analysis using Physiological Signals) is a widely used multi-modal dataset in emotion classification. (see TABLE 1 for a brief description): In the DEAP dataset, the channel no. 37 is for EDA signal and channel no. 38 is for RB signal. For our experiment, we have considered EDA and RB signals for the multi-modal physiological data stream.

3.2. Feature Extraction

It's important to extract features from different physiological signals (such as EDA, RB etc.) for retrieving relevant information, which effectively represents emotion states. Based on the extracted features the emotion classifier is tested and trained.

The most popularly and widely used time-frequency analysis of various signals (especially EDA, ECG etc.) for feature extraction is Wavelet Decomposition (WD) [**?**]. The reason of its popularity and widely use is because of localized analysis approach(i,e. time-frequency), multi-rate filtering, multi-scale zooming and it is better suited for non-stationary signals (such as EDA) [**?**]. Most frequently used wavelet base functions are Meyer WD, Morlet Mother WD, Haar Mother WD and Daubechies WD [**?**]. The most frequently used features extracted from each sub-bands of EDA and RB are entropy, median, mean, standard deviation, variance, 5th percentile value, 25th percentile value, 75th percentile value, 95th percentile value, root means square value, zero crossing rate, mean crossing rate [**? ? ?**]. In

[1]DEAP dataset link: `https://www.eecs.qmul.ac.uk/mmv/datasets/deap/`

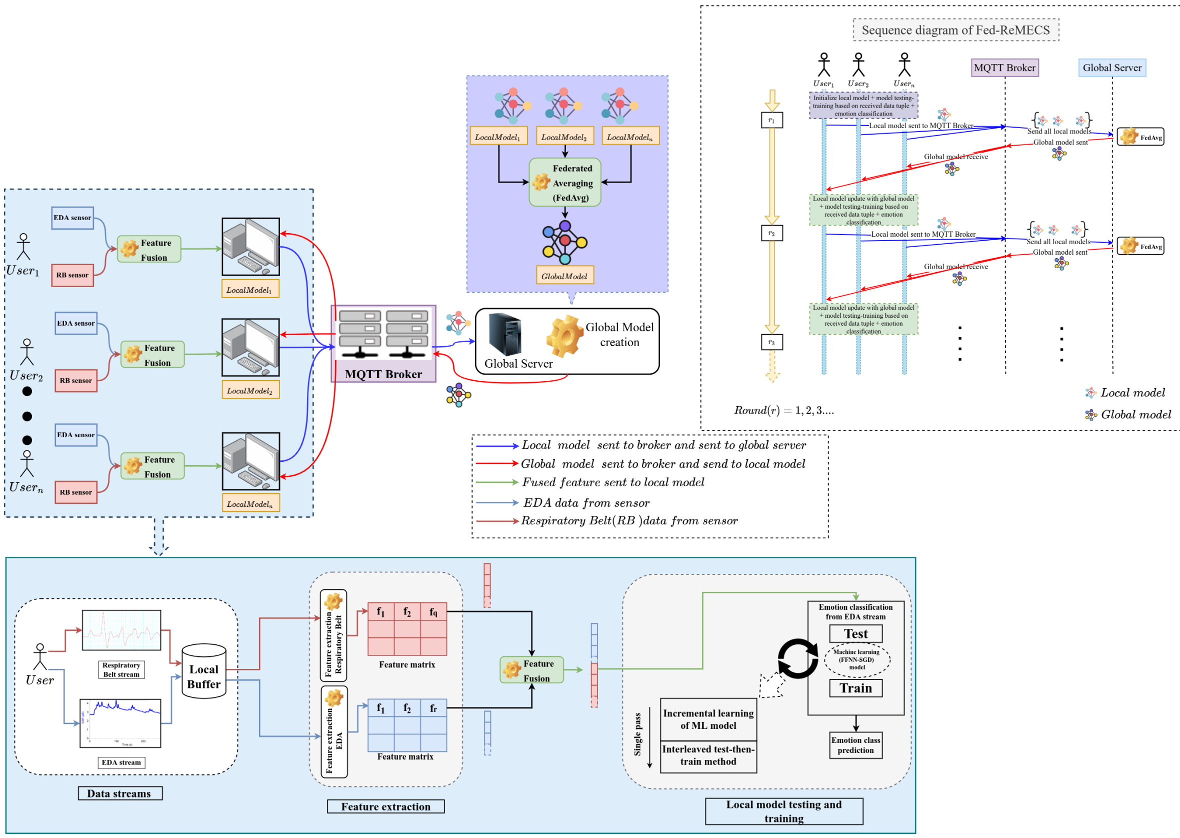

Figure 2: The Fed-ReMECS framework including the sequence diagram.

our experiment we have extracted and used these features for emotion classification.

3.3. Experimental Study

The whole experimental study of our proposed Fed-ReMECS approach in case of federated learning method for real-time emotion classification using multi-modal physiological data streams (such as EDA+RB data streams for DEAP dataset) is presented here.The Fed-ReMECS framework including the sequence diagram is depicted in Fig. 2.

There are three major parties are involved in Fed-ReMECS, one is at client's end, second one is the MQTT broker end and the final one is at global server end.

• **At client's end:** The clients are the end users having access to the generated data streams. Here at the client end, the major operations are (1) data stream reading from EDA and RB sensors, (2) feature extraction and feature fusion, (3)local model creation (testing and training) (4) emotion classification, (5) local model send and global model receive and (6) local model update. In our proposed approach, the clients acts as a *publisher* when the local model is created and ready to be sent to the MQTT broker; and also acts as a *subscriber* when it receives the global model from the MQTT broker send by global server.

• **At global server end:** The global server is responsible for the two functionalities (1) global model creation and (2) broadcasting the global model to clients. Global server creates the global model from all the local models from clients using the FedAvg approach. Once the global model is prepared, then it is sent to the MQTT broker and each client receives the global model. The global server acts as a *subscriber* when it is time to receive the local models and also acts as a *publisher* when it send the global model to the MQTT Broker.

• **At MQTT broker end:** The main functionality of the MQTT broker is (1) receive the local models from all clients and forward them to the global server; (2) send the global model to each client after the global model creation.

Now, each major operations at clients and global server end are described as follows:

1. *Data set consideration and data rearrangement:* The pre-processed multi-modal DEAP data [**?**] is used in this study to develop the multi-modal data stream. The first experiment is done using DEAP dataset where EDA and RB signal streams. As the DEAP is stored in 3D matrix representation, for easier EDA and RB data representation a rearrangement is done to convert the data into 1D matrix as follows:

$$[participant, video, data, valence\ class, arousal\ class]$$

In the experiment using DEAP dataset, the EDA and RB multi-modal data streams are used to classify

Table 2
Discrete emotion mapping with the help of valence–arousal status

Valence–Arousal level	Discrete emotions
LV-LA	sad (0)
LV-MA	Miserable (1)
LV-HA	Angry (2)
MV-LA	Tired (3)
MV-MA	Neutral (4)
MV-HA	Tense (5)
HV-LA	Calm (6)
HV-MA	Happy (7)
HV-HA	Excited (8)

discrete emotion states based on valence-arousal measurements. While streaming from DEAP dataset, an automatic mapping of the valence and arousal scores are done [**?**]; valence greater than equal to 1 and less than 4.5 is considered as *low valence (LV)*, valence greater than equal to 4.5 and less than 5.5 is considered as *mid valence (MV)*; and finally valence greater than equal to 5.5 and less than equal to 9 is considered as *high valence (HV)*. For arousal class labels similar scaling is done (such as *low arousal (LA)*, *mid arousal (MA)* and *high arousal (HA)*). This mapping is done because in DEAP dataset the valence-arousal measurements are available in $0-9$ range, according to the original DEAP paper [**?**]. Once the valence-arousal are mapped in these three level then the valence-arousal emotions are mapped into 9 discrete emotion labels with the help of the literature [**?**] and [**?**]. In TABLE 2 the discrete emotion labels are presented:

2. *Stream reading:* In the multi-modal data (DEAP) stream, we have used a sliding window protocol to stream the data for every participant. The sliding window size is set to 10s because it has already been used in previous emotion literature [**? ?**] and its effectiveness has been validated. The multi-modal data stream rate is approximately 3 Mb/10s for both the DEAP dataset. The Fed-ReMECS is developed using MQTT protocol.
3. *Feature extraction and fusion:* Wavelet feature extraction technique is used to extract features from multi-modal signal streams (EDA and RB signals from DEAP dataset) in this experiment. The wavelet Daubechies 4 (Db4) is the base function for feature extraction. In our experiment, decomposition of EDA and RB into three levels are performed. After that, the extracted features from EDA and RB modality a feature fusion approach (concatenation approach) is used. Then the fused features are sent to the emotion classifier in the client end.
4. *Emotion classifier:* To classify the discrete emotion labels (in TABLE 2), we have employed a three-layer Feed Forward Neural Network (FFNN) as the base classifier to do the classification in real-time from multi-modal data streams (EDA and RB). It is already been established in our previous work [**?**] that the FFNN is best classifier for real-time emotion classification from multi-modal physiological data streams. According to the TABLE 2, we have classified 9 discrete emotions.
5. *Local model test-train:* The FFNN models are trained with Stochastic Gradient Descent (SGD) in online fashion. The interleaved test-then-train [**?**] method is used for all considered base classifiers to evaluate because it uses same memory for testing and training; tests the model before it is trained and then the performance metrics are updated using the received data tuple. Thus, the base classifier is always tested on the received data tuples the classifier has never seen before. Initially, the base classifier performs very poorly but gradually it gains its stability and performance gets better as it sees more data tuples from multi-modal data streams.
6. *Local model send and global model receive:* The exact time to send the local model and receive the global model is completely problem dependent. In our experiment, we have considered the send and receive time according to the DEAP dataset experimental design. In the DEAP dataset each participant watch 60s video at a time, so the local model will be build on that time and after finishing each 60s video, the local model will be send to the MQTT broker and further to global server for global model creation. As soon as the global model is created it will be send to the MQTT broker and further sent to the clients for updating the local model with the received global model. Our consideration is that there is no communication delay in the creation of global model and deliver it to the clients.
7. *Global model creation:* The global server is responsible of creating the global model after performing Federated Averaging (FedAvg) on all the received local models. The FedAvg formula is as follows (in Eq. 1):

$$w_t^g = \frac{1}{N_{total}} \sum_{i=1}^{N_{total}} w_{t,i}^l \tag{1}$$

   Where w_t^g is the global model created at time t, N_{total} is the total number of local model received at global server, $w_{t,i}^l$ is the local model received from all clients at time t. In our scenario, we have considered full participation from all clients in FedAvg.
8. *local model update:* When each client receives the global model, their local model is updated with the global model and then each client starts for the next federated learning iteration. Each local models weight are updated as follows (in Eq: 2):

$$w_{t+1,i}^l \leftarrow w_t^g - \lambda \nabla_{w_t^g} L(w_t^g) \tag{2}$$

   Where L is the loss function of the local model (FFNN), $\nabla_{w_t^g}$ is the local model gradient of each clients and λ is the learning rate. In Fed-ReMECS, the categorical cross-entropy loss function is used. The mathematical formula of categorical cross-entropy loss function is in Eq. 3

$$L = -\sum_{i=1}^{N} y_c^o \log(p_c^o) \tag{3}$$

Where N is the number of classes (in our case, total 9 emotion class labels), y is the binary indicator (0 or 1) of the class label c for the observation o and p is the predicted probability of observation o is of class c.

3.4. Experimental Setup

Herewith the machine setup, software development, experiment environment and parameter setup are as follows:

• *Machine configuration:* Ubuntu 18.04 64 bit OS, processor core-i7-7700HQ with RAM 24 Gb–2400MHz and 4Gb-Nvidia GTX-1050 graphics.

• *Software development:* The Fed-ReMECS is implemented in Python 3.7 with the help of TensorFlow[2] and Keras[3] in the backend. For the Subscriber and Publisher implementation we have used paho-mqtt[4] library. Also, for the MQTT broker the Eclipse Mosquitto broker[5] is installed in our local computer.

• *Parameter setup:* The learning rate for SGD is 0.05 for local model FFNN and it is fixed in the whole emotion classification. For the hidden layer *sigmoid* activation function and for the output layer *softmax* is used. The *epoch* is set to 1 because the model observes the data once, which implies that model training happens once for each incoming data tuple.

• *Emotion classifier visualization:* The emotion classifier (3-layer FFNN) used in Fed-ReMECS is depicted in Fig. 3.

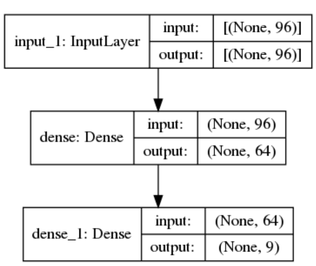

Figure 3: The pictorial view of the 3-layer FFNN used for emotion classification in Fed-ReMECS.

• *Source-code:* The source code and implementation details of our developed Fed-ReMECS can be found in *Github*[6].

3.5. Performance Metrics

For evaluating the classifiers performance, accuracy (Acc) [**?**] is used. These metrics are calculated as follows:

[2] https://www.tensorflow.org/

[3] https://keras.io/

[4] https://pypi.org/project/paho-mqtt/

[5] https://mosquitto.org/download/

[6] The source code can be found in GitHub: https://github.com/officialarijit/Fed-ReMECS-mqtt

Table 3
Fed-ReMECS performance in terms of average accuracy

No. of clients	Accuracy
5	0.8384 (±0.29)
10	0.8423 (±0.30)
15	0.8205 (±0.33)
20	0.8384 (±0.28)
25	0.8263 (±0.26)
32	0.8192 (±0.21)

$$Acc = \frac{\sum_{i=1}^{|l|} \frac{TP_i + TN_i}{TP_i + FN_i + FP_i + TN_i}}{|l|} \tag{4}$$

where $|l|$ is the number of classes. True positives (TP_i), True negatives (TN_i), False positives (FP_i) and False negatives (FN_i).

4. Results, Analysis and Discussion

In this we have presented the experimented results of our developed Fed-ReMECS system under different number of clients running in parallel. To test the effectiveness of Fed-ReMECS, we have considered total 6 number of clients setting (5, 10, 15, 20, 25 and 32). It means first experiment is conducted by considering 5 clients running in parallel, second experiment is 10 clients are running parallel and so on. Running client in parallel means, reading and sending the data from the DEAP dataset for emotion classification using ReMECS approach , because each client is running ReMECS in their own system. The average accuracy of the global model is shown in TABLE 3 and the performance change in terms of accuracy is shown in Fig. 4.

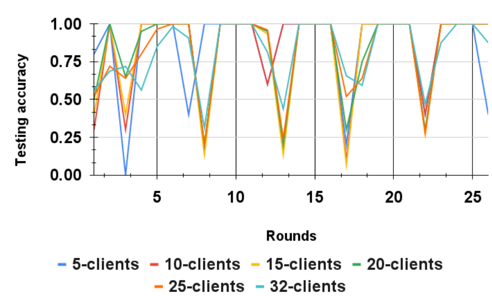

Figure 4: Global model's accuracy in Fed-ReMECS for different number client setting

As, in Fed-ReMECS the real-time emotion classification is done by each clients local model because data accessibility is strictly restricted to each clients, there is no way the global server gets access to the data. That's why, the accuracy of the global model is calculated by taking the average of each local model's first accuracy value after receiving the global model from global server. Now, from the average accuracy (in TABLE 3), it can be observed that the Fed-ReMECS approach is capable of performing good in real-time emotion classification while maintaining adequate accuracy. Also, in different parallel client setting the effectiveness and scalability of the Fed-ReMECS can also be worth noticing.

From Fig. 4, we can observe that the global model's performance has changed over the period (different rounds) because the model updates itself gradually and hence the performance change but ultimately showed good overall accuracy. However, in some periods, the model's accuracy has dropped because the performance of real-time emotion classification at some clients' end has dropped. In federated learning, as the global model have no access to the data, thus the accuracy is calculated by averaging all the local models' accuracy. So, dropping one local model's performance can decrease the overall global model's performance.

Finally, based on the average accuracy and the overall performance results we can conclude that the Fed-ReMECS system has the capability of classifying the emotions in real-time from multi-modal physiological data streams without accessing the actual sensitive multi-modal data streams. So, Fed-ReMECS is capable of preserving the privacy issue by building a power enough global emotion classifier from local model parameters and without accessing the sensitive user data. Having said that, the experiment conducted using different number of client setting, Fed-ReMECS can handle multiple clients at the same time running in parallel. Hence Fed-ReMECS is scalable and capable for large-scale practical adaptation.

5. Application Scenario

Our developed Fed-ReMECS system has potential real-life application scenarios, described as follows:

- *In-home health monitoring:* In-home health monitoring has received a lot of interest from the world's aging population. With the abundance of user health data acquired by Internet of Things (IoT) devices and recent advancements in machine learning, smart healthcare has seen several success stories. IoT devices, particularly wearable devices such as smartphones and smartwatches, may continually track a user's activity, heart rate, and so on, and then send this information to a healthcare center for further analysis and diagnosis. Smart in-home healthcare is able to support ubiquitous and non-intrusive health monitoring because to the ease and cheap cost of acquiring people's health information, and is developing as a mainstream for smart healthcare.

- *Workers in factory:* Excessive stress is harmful to one's health. When workers in a factory are in a high-stress situation, they quit their employment nearly three times more likely, inhibits strategic thinking for a short period of time, and dulls creative ability. If the workers stay in this stressed mode too long, they get burned out. To counteract this impact, we must create a secure work environment and implement stress-reduction practices into your team's regular operations. So, to create a stress free work environment real-time stress level detection of the workers is necessary.

- *Students in online learning:* Emotion state recognition of the students and responding to the emotion states is very crucial. In conventional classroom teaching it is successful, but very difficult in the scenario when students are learning in online mode. In the online-learning environment, it is not just learning but also the interdependence between learning and emotion that is mediated (teachers, students and learning). From the perspective of the learners, the anticipated positive outcomes are appropriate decision making, learning activity management, and time management; integrating all of these outcomes will increase students' attention, motivation, and desire to study. So, to accomplish the above mentioned outcomes, its necessary to recognize emotion states in real-time while students are learning online.

Existing systems for the above mentioned application scenarios do not pay enough attention to user data privacy and are therefore far from ready for large-scale practical adoption. Our developed Fed-ReMECS system can easily fulfill the data privacy concern and large-scale practical adaptation. Because in Fed-ReMECS, no data in exchanged in global server and clients only the model prepared using the data only accessible to the clients, which addresses and solves the data privacy issue. Also, Fed-ReMECS is capable for the large-scale practical adaptation, because it is developed using the light weight MQTT protocol and the data is processed in real-time for emotion state classification; hence large-scale adaptation can be handled.

6. Conclusion

This paper presents Fed-ReMECS, a federated learning framework for real-time emotion state classification from multi-modal data streaming. The data streams are generated at a high rate from various users' physiological measurements (EDA, RB, etc.). The key aspect of our proposed Fed-ReMECS is the ability to build a highly accurate global classifier without accessing distributed multi-data streams while identifying emotional states in real-time. In addition, the Fed-ReMECS technique can protect users' sensitive data privacy. Furthermore, the proposed framework is also suited for usage in an IoT environment because of the lightweight MQTT protocol. The experimental results based on the popular DEAP dataset showed that the Fed-ReMECS achieves high accuracy, is scalable, and has a great potential for practical adaption for emotional health application scenarios.

Future works include investigating different time embedding to increase the model's performance. Also, as this is a streaming scenario, concept drift may happen. So in that scenario, how does the model captures concept drift and adapt itself? Apart from this, we will further investigate the performance of Fed-ReMECS for some more focused application scenarios such as the track of users' vitals, etc.

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CRediT authorship contribution statement

Arijit Nandi: Conceptualization, methodology, software validation, formal analysis, investigation, and resources.

Fatos Xhafa: Conceptualization, methodology, software, validation, formal analysis, investigation, resources and supervision.

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